SUPPORTING ADULT GUIDANCE

Equations with fractions

Equations, Inequalities and Simultaneous Relationships · Lesson 3. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson extends equation-solving to equations containing fractions. The organising move is to **clear the fraction first** by multiplying every term by the denominator (or the lowest common multiple of the denominators), reducing the problem to the linear equations of the previous lessons. Checking by substitution remains the final step.

Driving Question: *How do I clear a fraction and keep the equation true?*

The mathematics, in plain terms

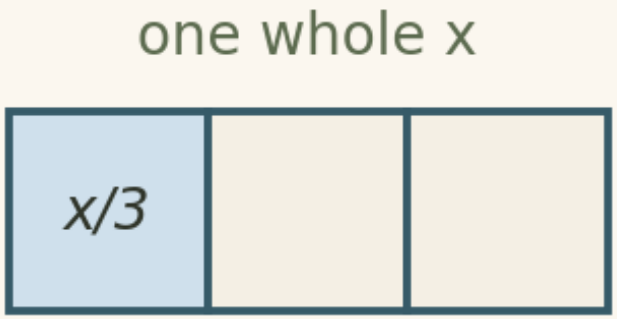
A term x/n is one nth of x; multiplying it by n returns x. Multiplying every term of an equation by n (a non-zero number) preserves the equality, so it is a legitimate move — and it clears a denominator of n. When two denominators are present, multiplying by their lowest common multiple clears both simultaneously. After clearing, the equation is an ordinary linear one solved by inverse operations, and substitution verifies the solution.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — A single fraction

Solve $x/3 + 2 = 5$. The bar shows $x/3$ — one of three equal parts of x.

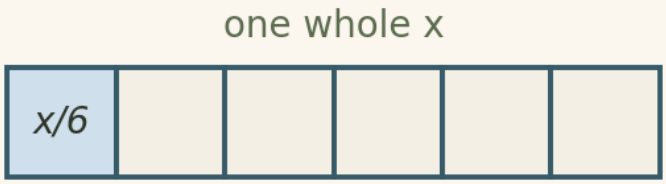


1. Multiply every term by 3 to clear the fraction.	$3\times(x/3) + 3\times2 = 3\times5$
2. The fraction is gone.	$x + 6 = 15$
3. Subtract 6 from both sides.	$x = 9$
4. Check in the original.	$9/3 + 2 = 3 + 2 = 5$ ✓

$x = 9$ — multiply every term by 3 to get $x + 6 = 15$, then subtract 6; the check gives 5.

Example 2 — Two denominators — use the LCM

Solve $x/2 + x/3 = 10$. The denominators are 2 and 3; their lowest common multiple is 6.



1. Multiply every term by 6 (the LCM of 2 and 3).	$6\times(x/2) + 6\times(x/3) = 6\times10$
2. Both fractions clear at once.	$3x + 2x = 60$
3. Collect like terms.	$5x = 60$
4. Divide both sides by 5.	$x = 12$
5. Check in the original.	$12/2 + 12/3 = 6 + 4 = 10$ ✓

$x = 12$ — the LCM of 2 and 3 is 6; multiplying through gives $3x + 2x = 60$, so $5x = 60$ and $x = 12$.

Clear the fraction first, then keep the balance

Encourage a fixed order: clear the fraction (multiply every term by the denominator or the LCM), then expand or collect if needed, then clear the number, then divide, then check. The single most common error is multiplying only some terms — stress *every* term, both sides. A learner who clears first meets only equations they can already solve.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	See x/n as one of n equal parts; n copies rebuild the whole x.	~8 min
Movement	Clear a fraction by multiplying every term by the denominator.	~11 min
Reflection	The multiply-through method written down, line by line, then checked.	~12 min
Creativity	Use the LCM to clear two different denominators at once.	~9 min
Rest	A short pause on making the awkward-looking simple.	~2 min
Nutrition	Name what the learning feeds: composure with the awkward-looking.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Multiplying only the fraction, not every term: $x/3 + 2 = 5 \rightarrow x + 2 = 5$.
Return to 'do the same to the whole side': 'If you multiply the left by 3, every term on the left is multiplied — what is 3 lots of 2? And the right side too.'

Forgetting the other side entirely.
The equation only stays true if both sides are multiplied. 'You multiplied the left by 3 — what must happen to the 5?'

With two denominators, multiplying by just one of them.
One multiplier must clear both. 'What is the smallest number both 2 and 3 divide into? Multiply every term by that.'

Solving but never checking.
Make substitution routine: 'Put your value back into the original — do both sides come out equal?'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: fractions of a quantity, the lowest common multiple, and keeping an equation balanced. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner writes $x/3 + 2 = 5 \rightarrow x + 2 = 5$ (multiplying only the fraction), or clears one denominator of two. The aim is a question that hands the reasoning back.

- “If you multiply the left side by 3, how many terms get multiplied — just the fraction, or all of them?”
- “What is 3 lots of 2? And you must do the same to the 5 on the right.”
- “With x/2 and x/3, what is the smallest number both 2 and 3 divide into?”
- “Put your answer back into the original — do both sides come out the same?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Multiply by the denominator, or divide?

Multiply every term by the denominator — that is what removes it. Dividing would make more fractions, not fewer.

What if there are two different denominators?

Multiply every term by their lowest common multiple; it clears both fractions in one step. Listing a few multiples of each denominator finds the LCM quickly.

Do the answers always come out whole?

Not always — the stretch question gives x = 2.5. Fractions and decimals are perfectly good answers; the check works the same way.

How does this build on the earlier lessons?

Once the fraction is cleared, every equation here is one from Lessons 1 and 2. The new skill is the clearing step — multiplying through — not a new way of solving.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like

Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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